

Lee, Elisa, T. (1992): Statistical Methods for Survival Data Analysis, John Wiley & Sons.

Li, C.C. (1976): First Course of Population Genetics, Boxwood Press.

Fisher, L.D. and Belle, G.V. (1993): Biostatistics: A Methodology for the Health Science, John Wiley & Sons Inc.

Lawless, J.F.(2003): Statistical Methods for Lifetime (Second Edition), John Wiley & Sons.

Rosner B. (2006): Fundamentals of Biostatistics, Edition 6.

rogramme	B. Sc. Statistics				
Course Code	STA5EJ304				
Course Title	Econometrics				
Type of Course	Major Elective				
Semester	V				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites					
Course Summary Objective	After completing the course students should be able to interpret regression results as well as to understand the assumptions underlying the ordinary least squares estimator, and judge in an educated manner whether they hold in a given problem.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	will be able to estimate and interpret linear regression models and be able to distinguish between economic and statistical importance.	U	C	Instructor-created exams / Quiz
CO2	They should be able to use a statistical/econometric computer package to estimate an econometric model	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	To understand the scope and application of econometrics to real world problems.	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	To know econometric problems and their solutions	U	C	Instructor-created exams / Home Assignments
CO5	Student also will be exposed to simple statistical packages and their use in econometric work	Ap	P	One Minute Reflection Writing assignments
CO6		Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I		Introduction	10	15
	1	Purpose and scope of econometrics		
	2	Econometric model		
	3	Model building and role of econometrics.		
	4	General linear model (GLM).		
	5	Estimation under linear restrictions and properties of estimators		
	Sections from References:			

II	Heteroscedasticity		12	20
	6	Econometric problems		
	7	Heteroscedasticity		
	8	Tests for heteroscedasticity,		
	9	Consequences of heteroscedasticity and solutions		
	Sections from References:			
III	Autocorrelation		12	15
	10	Autocorrelation concept		
	11	Consequences of auto correlated disturbances,		
	12	Detection of Autocorrelation		
	13	Tests of autocorrelation.		
	14	Distributed lag models		
	15	Estimation of parameters		
	Sections from References:			
IV	Multiple regression		14	20
	16	Concept of Multiple regression		
	17	Multiple regression analysis.		
	18	Multi collinearity: Introduction and concepts,		
	19	Detection of multi collinearity,		
	20	Consequences multi collinearity		
	21	Sources multi collinearity		
	22	Tests and estimation of multi collinearity		
	Sections from References:			
V	Open Ended Module:		12	
	1	Practical Problems related to OLS/ CLR using softwares. Introduction to various Economic functions (Demand , Supply, Utility, Cost , Revenue etc.)		
	Sections from References:			
Books and References:				
1. Gujarathi, D. and Sangeetha, S.(2007). Basic Econometrics, Mc Graw Hill				